

Ubuntu va OS asoslari: to'liq mashqlar

P1-P6: qadamma-qadam

Ubuntu Server 24.04 LTS / ochiq Lab v1.0.0
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Ubuntu and OS Foundations — to'liq ko'rsatmalar bilan mashqlar

Qamrov: Guided Practice P1–P6. Muhit: Ubuntu Server 24.04 LTS. Har bir P mashqidan keyin shu ko'nikma bilan Mission Mni mustaqil bajaring.

0. Umumiy qoidalar

P1–P6 mashq uchun. Ular M1–M6dan boshqa `directory`, `user`, `group`, `service` va `port`lardan foydalanadi. M7 umumlashtiruvchi topshiriq; unga alohida qadamma-qadam mashq yo'q.

Ubuntuda quyidagini bajaring. Natija `ssm-user` bo'lsin:

```
id -un
```

Boshqa `user` ko'rinsa, bir marta `exit` qiling va qayta tekshiring:

```
exit  
id -un
```

Har bir bosqich boshida ish `directory`siga o'ting. O'tgach `pwd` bilan joyni tekshiring.

Baholash:

```
jdu-check P1
```

Faqat boshidan boshlash zarur bo'lsa `reset` qiling:

```
jdu-reset P1
```

P1 Shell, path, file, text

O'rganiladigan ko'nikmalar

- `home directory` va ish `directory`sini ajratish.
- `directory` yaratish.
- `file`dan nusxa olish.
- `log`dan kerakli satrlarni saqlash.

1-bosqich: Home directoryga o'tish

```
cd ~  
pwd
```

Natija `/home/ssm-user` bo'lsin.

2-bosqich: P1 directorysiga o'tish

```
cd ~/jdu-lab/p1
pwd
tree
```

inbox va hali tugallanmagan practice01ni toping.

3-bosqich: Kerakli directorylarni yaratish

```
cd ~/jdu-lab/p1
mkdir -p practice01/config
mkdir -p practice01/logs
mkdir -p practice01/notes
tree practice01
```

4-bosqich: Config filedan nusxa olish

Nusxa joylashtiriladigan directoryga o'ting.

```
cd ~/jdu-lab/p1/practice01/config
pwd
cp .././inbox/config/training.conf .
ls -l
```

. hozirgi directoryni anglatadi.

5-bosqich: Log filedan nusxa olish

```
cd ~/jdu-lab/p1/practice01/logs
pwd
cp .././inbox/logs/practice.log .
ls -l
```

6-bosqich: Vaqtinchalik filelarni topib o'chirish

practice01ga o'ting:

```
cd ~/jdu-lab/p1/practice01
pwd
find . -type f -name '*.tmp' -print
find . -type f -name '*.tmp' -delete
find . -type f -name '*.tmp' -print
```

Oxirgi command hech narsa chiqarmasligi kerak.

7-bosqich: WARN bor satrlarni saqlash

Natija yoziladigan notesga o'ting:

```
cd ~/jdu-lab/p1/practice01/notes
pwd
grep 'WARN' ../logs/practice.log > warnings.txt
cat warnings.txt
```

8-bosqich: Oxirgi to'rt satrni saqlash

```
cd ~/jdu-lab/p1/practice01/notes
tail -n 4 ../logs/practice.log > recent.txt
cat recent.txt
```

9-bosqich: Barcha natijani tekshirish

```
cd ~/jdu-lab/p1
tree practice01
stat -c '%U:%G %a %n' practice01 practice01/config/training.conf practice01/logs/practice.log practice01/
notes/warnings.txt practice01/notes/recent.txt
jdu-check P1
```

Barcha 6 band **PASS** bo'lsa, M1ga o'ting.

P2 User, group, permission, setgid

O'rganiladigan ko'nikmalar

- **primary group** va **supplementary group**ni tekshirish.
- Umumiy directoryga **group permission** qo'yish.
- **setgid** yordamida yangi **filening group**ini meros qildirish.

1-bosqich: Boshqaruv userini tekshirish

```
cd ~
id -un
id
```

2-bosqich: Boshlang'ich groupni tekshirish

```
getent group practiceops
id jdupracticewriter
id jdupracticeviewer
```

Dastlab **jdupracticeviewer** **practiceops** a'zosi, **jdupracticewriter** esa a'zo emas.

3-bosqich: Group a'ziligini to'g'rilash

```
cd ~
sudo usermod -aG practiceops jdupracticewriter
sudo gpasswd -d jdupracticeviewer practiceops
getent group practiceops
id jdupracticewriter
id jdupracticeviewer
```

4-bosqich: Umumiy directoryni sozlash

```
cd /srv
pwd
sudo chown root:practiceops jdu-practice-share
sudo chmod 2775 jdu-practice-share
stat -c '%U:%G %a %n' jdu-practice-share
```

5-bosqich: GUIDE.txtni sozlash

Kerakli directoryga o'ting:

```
cd /srv/jdu-practice-share
pwd
sudo chown root:practiceops GUIDE.txt
sudo chmod 664 GUIDE.txt
stat -c '%U:%G %a %n' GUIDE.txt
```

6-bosqich: Writerga o'tib file yaratish

```
cd ~
sudo su - jdupracticewriter
id -un
id
cd /srv/jdu-practice-share
pwd
printf '%s\n' 'guided writer file' > writer-created.txt
ls -l writer-created.txt
exit
id -un
```

Oxirgi natija ssm-user bo'lsin.

7-bosqich: Viewerning o'qish va yozishini tekshirish

```
cd ~
sudo su - jdupracticewriter
id -un
id
cd /srv/jdu-practice-share
pwd
cat GUIDE.txt
touch viewer-created.txt
exit
id -un
```

cat muvaffaqiyatli ishlaydi. touch Permission denied beradi. Bu kutilgan rad etish.

8-bosqich: Baholash

```
cd ~
id -un
jdu-check P2
```

Barcha 6 band PASS bo'lsa, M2ga o'ting.

P3 Process va package

8-bobdan keyin 1–4-bosqichlar orqali processni tekshirib to'xtating. 9-bobni o'qib, 5–6-bosqichlar orqali packageni o'rnatish. Ikkala qism tugagach jdu-check P3ni bajaring.

O'rganiladigan ko'nikmalar

- systemd service va process PIDini bog'lash.
- Aniq PIDga TERM signal yuborish.
- apt bilan package o'rnatish.

1-bosqich: Uchta serviceni tekshirish

```
cd ~
systemctl list-units --type=service 'jdu-p3-*'
systemctl status jdu-p3-process1.service --no-pager
systemctl status jdu-p3-process2.service --no-pager
systemctl status jdu-p3-process3.service --no-pager
```

2-bosqich: Process2 Main PIDini ko'rsatish

```
systemctl show --property MainPID --value jdu-p3-process2.service
```

3-bosqich: PIDni shell o'zgaruvchisida saqlash

```
P3_PID=$(systemctl show --property MainPID --value jdu-p3-process2.service)
printf '%s\n' "$P3_PID"
ps -fp "$P3_PID"
```

4-bosqich: Faqat process2ga TERM yuborish

```
sudo kill -TERM "$P3_PID"
systemctl is-active jdu-p3-process1.service
systemctl is-active jdu-p3-process2.service
systemctl is-active jdu-p3-process3.service
```

Tartib bilan active, inactive, active chiqishi kerak.

5-bosqich: Figletni tekshirib o'rnatish

```
cd ~
apt show figlet
sudo apt update
sudo apt install -y figlet
command -v figlet
figlet JDU
```

6-bosqich: Package va command mosligini ko'rish

```
dpkg-query -W -f='${Status} ${Version}\n' figlet
dpkg -S /usr/bin/figlet
jdu-check P3
```

Ikkala band `PASS` bo'lsa, `M3ga` o'ting.

P4 systemd service

O'rganiladigan ko'nikmalar

- `unit` fileni o'qish.
- `active` va `enabled` holatlarini alohida sozlash.
- `Main PID` va haqiqiy `process`ni bog'lash.

1-bosqich: Unit fileni o'qish

```
cd ~
systemctl cat jdu-practice-status.service
```

`User`, `WorkingDirectory`, `ExecStart`ni tekshiring.

2-bosqich: Dastlabki holatni tekshirish

```
systemctl is-active jdu-practice-status.service
systemctl is-enabled jdu-practice-status.service
```

3-bosqich: Serviseni start qilish

```
sudo systemctl start jdu-practice-status.service
systemctl is-active jdu-practice-status.service
```

4-bosqich: Tizim yoqilganda start qilishni sozlash

```
sudo systemctl enable jdu-practice-status.service
systemctl is-enabled jdu-practice-status.service
```

5-bosqich: Main PID va processni tekshirish

```
P4_PID=$(systemctl show --property MainPID --value jdu-practice-status.service)
printf '%s\n' "$P4_PID"
ps -fp "$P4_PID"
sudo cat "/proc/$P4_PID/cmdline" | tr '\0' ' '
printf '\n'
sudo readlink -f "/proc/$P4_PID/cwd"
```

6-bosqich: Baholash

```
cd ~  
jdu-check P4
```

Uchala band `PASS` bo'lsa, M4ga o'ting.

P5 Port, socket, HTTP, journal

O'rganiladigan ko'nikmalar

- `service`, `process` va `listening socket`ni bog'lash.
- IP address va portni o'qish.
- HTTP request `journal`da qayd etilishini tekshirish.

1-bosqich: Unit fileni o'qish

```
cd ~  
systemctl cat jdu-practice-web.service
```

127.0.0.1, 8181, `jdupracticeweb` va `kontent` filei pathini tekshiring.

2-bosqich: Serviceni start qilish

```
sudo systemctl start jdu-practice-web.service  
systemctl is-active jdu-practice-web.service
```

3-bosqich: Listening socketni tekshirish

```
sudo ss -ltnp | grep ':8181'
```

127.0.0.1:8181ni toping. Shu chiqishda 0.0.0.0:8181 yoki :::8181da kutish yo'qligini ham tekshiring. Birinchisi faqat shu host ichidan ulanishni, keyingilari barcha manzillarda kutishni bildiradi.

4-bosqich: Socket PID va Main PIDni solishtirish

```
systemctl show --property MainPID --value jdu-practice-web.service  
sudo ss -ltnp | grep ':8181'
```

Ikki PID bir xil bo'lishi kerak.

5-bosqich: HTTP responseni tekshirish

```
curl -i http://127.0.0.1:8181/
```

6-bosqich: Service userning o'qish huquqini tekshirish

Kontent `directory`siga o'ting:

```
cd /srv/jdu-practice-web
pwd
stat -c '%U:%G %a %n' index.txt
namei -l /srv/jdu-practice-web/index.txt
id jdupracticeweb
sudo -u jdupracticeweb -- test -r index.txt
echo $?
```

Oxirgi qiymat 0 bo'lsa, o'qish mumkin.

7-bosqich: Talaba requestini yuborish

```
curl -i http://127.0.0.1:8181/p5-check
sudo journalctl -u jdu-practice-web.service --no-pager -n 20
```

REQUEST path=/p5-checkni toping.

8-bosqich: Closed port bilan solishtirish

```
curl --max-time 2 http://127.0.0.1:18181/
sudo ss -lnt | grep ':18181'
```

Ikkalasi ham muvaffaqiyatli chiqmaydi. TCP 18181da kutayotgan socket yo'q.

9-bosqich: Baholash

```
cd ~
jdu-check P5
```

To'rttala band PASS bo'lsa, M5ga o'ting.

P6 SSH, remote command, scp

O'rganiladigan ko'nikmalar

- CloudShell va Ubuntuni farqlash.
- scp bilan upload va download qilish.
- SSH remote commandini bajarish.

1-bosqich: Ubuntudan CloudShellga qaytish

Ubuntu promptida bajaring:

```
exit
```

CloudShellda qayerda ekaningizni tekshiring:

```
cd ~
pwd
id -un
hostname
```

2-bosqich: CloudShelldagi ish directoriesini yaratish

```
mkdir -p ~/jdu-lab/p6  
cd ~/jdu-lab/p6  
pwd
```

3-bosqich: Upload manbasi bo'lgan fileni yaratish

```
printf '%s\n' 'JDU SSH guided transfer' > practice-source.txt  
cat practice-source.txt  
sha256sum practice-source.txt
```

4-bosqich: Ubuntu'dagi directoryni yaratish

CloudShellda bajaring:

```
ssh jdu-ubuntu 'mkdir -p ~/jdu-lab/p6'
```

5-bosqich: Ubuntu'ga upload qilish

CloudShell P6 directorysidan bajaring:

```
cd ~/jdu-lab/p6  
scp practice-source.txt jdu-ubuntu:~/jdu-lab/p6/practice-upload.txt  
ssh jdu-ubuntu 'sha256sum ~/jdu-lab/p6/practice-upload.txt'  
sha256sum practice-source.txt
```

Ikki SHA-256 qiymati bir xil bo'lsin.

6-bosqich: SSH remote command bilan natija fileni yaratish

CloudShellda quyidagi bitta commandni aynan ko'rsatilgandek bajaring:

```
ssh jdu-ubuntu 'cd ~/jdu-lab/p6 && printf "REMOTE_USER=%s\nREMOTE_HOST=%s\nREMOTE_PATH=%s\n" "$(id -un)"  
"$(hostname)" "$HOME" > practice-remote-result.txt'
```

Mazmunni remoteda tekshiring:

```
ssh jdu-ubuntu 'cat ~/jdu-lab/p6/practice-remote-result.txt'
```

7-bosqich: Ubuntu tomonini baholash

Ubuntu'ga ulanib bajaring:

```
ssh jdu-ubuntu  
cd ~/jdu-lab/p6  
pwd  
ls -l  
jdu-check P6  
exit
```

Ubuntu tomondagi ikkala band PASS bo'lsin.

8-bosqich: Natija fileni download qilish

CloudShellda bajaring:

```
cd ~/jdu-lab/p6
scp jdu-ubuntu:~/jdu-lab/p6/practice-remote-result.txt practice-downloaded-result.txt
cat practice-downloaded-result.txt
sha256sum practice-downloaded-result.txt
ssh jdu-ubuntu 'sha256sum ~/jdu-lab/p6/practice-remote-result.txt'
```

Ikki SHA-256 qiymati bir xil bo'lsin.

9-bosqich: CloudShell tomonini baholash

```
cd ~/jdu-lab/p6
jdu-check P6
```

CloudShell tomondagi to'rtala band **PASS** bo'lsa, M6ga o'ting.

O'rganish tartibi

Kitobdagi boblar dars soatlari bilan birma-bir mos emas. Tushungan joyingizdan mashqqa o'ting, zarur bo'lsa bobga qayting.

- 1–3-boblar va 4-bobning tahrirlash asoslarini o'qing. Keyin M0da haqiqiy OSni kuzating.
- 4–5-boblarni o'qib P1, M1ga o'ting.
- 6–7-boblarni o'qib P2, M2ga o'ting.
- 8-bobni o'qib P3ning 1–4-bosqichlarini bajaring. 9-bobni o'qib P3ning 5–6-bosqichlarini bajaring. Keyin M3ga o'ting.
- 10-bobdan keyin P4, M4; 11-bobdan keyin P5, M5ga o'ting.
- 12-bobdan keyin P6, M6ga o'ting. Avvalgi ko'nikmalarni birlashtirib M7ni bajaring.

Umumiy tavsiya etilgan tartib:

```
P1 → M1 → P2 → M2 → P3 → M3 → P4 → M4
→ P5 → M5 → P6 → M6 → M7 (yakuniy topshiriq)
```

Pda ko'rsatilgan **command**larga qarab ishleng. M1–M6da Pdagi **command**larga qaramasdan, keraklisini o'zingiz tanlang. M7da M2, M4 va M5dagi bilimlarni birlashtiring.